

BUSINESS REQUIREMENTS DOCUMENT

Social Group Buying E-Commerce Platform

Multi-Vendor | Group Discounts | WhatsApp-Powered Social Commerce

Prepared For	Prepared By
Sahil Srivastava (Client)	ArgosMob Tech & AI Pvt Ltd Greater Noida West
Project Value	Document Date
INR 70,000 + 18% GST	April 2026
Timeline	Document Status
20-25 Business Days	Final

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1 Project Overview

Project Title	Social Group Buying E-Commerce Platform
Client Name	Sahil Srivastava
Developed By	ArgosMob Tech & AI Pvt Ltd, Greater Noida West
Document Date	April 2026
Project Type	Multi-Vendor Web Application
Project Value	INR 70,000 + 18% GST
Estimated Timeline	20-25 Business Days
Document Version	v1.0 – Final

1.1 Executive Summary

This Business Requirements Document (BRD) outlines the complete functional and technical scope for a Social Group Buying E-Commerce Platform commissioned by Sahil Srivastava and to be delivered by ArgosMob Tech & AI Pvt Ltd. The platform enables users to form buying groups for premium products, share invite links via WhatsApp, and unlock tiered discounts as more members join. Payments are only captured once a group successfully reaches its target size, ensuring a zero-risk experience for the buyer.

The platform follows a multi-vendor marketplace model, where independent sellers can list products, manage orders, and add shipping tracking details. An Admin Panel provides full control over products, sellers, group deals, and orders — including the ability to manually resolve incomplete or stalled groups.

The product draws direct inspiration from leading group-buy commerce models observed in platforms like Pinduoduo and Temu, adapted for the Indian e-commerce context with UPI-first payment infrastructure and WhatsApp as the primary social sharing channel.

2 Business Objectives

- 1

Enable Social Commerce
Allow users to form groups with friends and family to unlock better prices on premium products.
- 2

Reduce Cart Abandonment
Pre-authorization model charges only on group completion, reducing hesitation and abandoned checkouts.
- 3

Drive Virality via WhatsApp
Frictionless WhatsApp sharing turns every buyer into a distributor, lowering CAC organically.

4

Empower Independent Sellers

Provide sellers a dedicated panel to list products, manage inventory, and handle fulfillment independently.

5

Scalable Discount Engine

Tiered pricing (2, 3, 5, 10 members) ensures sellers can participate at their own margin comfort.

6

Admin Oversight & Control

Centralized admin to monitor deals, manage sellers, and intervene in stalled groups.

3

Scope of Work**3.1 In Scope**

- Responsive web application (desktop & mobile browser)
- Multi-vendor marketplace: seller product listings with commission model
- Group buying engine: group creation, invitation, real-time member tracking
- Tiered pricing tiers: Solo, 2, 3, 5, and 10 member discounts
- 24-hour group countdown timer with single-use extension option
- WhatsApp-native sharing with pre-filled invite messages and group links
- Payment gateway integration: UPI, Credit/Debit Card, Cash on Delivery
- Pre-authorization payment hold — charge only on group completion
- Auto-release of payment hold if group expires without meeting target
- Seller Panel: product listing, inventory, order management, tracking
- Admin Panel: full platform management, manual group completion capability
- Push + WhatsApp notifications for join events, completion, and expiry alerts
- Order management, shipping tracking, and delivery status updates
- Basic SEO configuration for product pages

3.2 Out of Scope

- Native mobile apps (iOS / Android) — web-responsive only
- Custom logistics or last-mile delivery infrastructure
- Third-party API costs (WhatsApp Business API, payment gateway charges)
- Content creation: product photography, copywriting, catalogue data entry
- Post-delivery warranty, return processing, or dispute arbitration workflows
- Integration with GST filing software or accounting platforms
- Machine learning-powered product recommendation engine
- Multi-language or regional language support

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Functional Requirements

4.1 User Roles & Access

Role	Description	Access Level
Guest / Visitor	Browse products, view pricing, join group via invite link	Read-only
Registered User	Create/join groups, manage cart, make payments, track orders	Standard
Seller	List products, set tier pricing, manage inventory & orders	Seller Panel
Admin	Full platform access, manual interventions, seller approvals	Admin Panel

4.2 Group Buying Engine

The group buying engine is the core value driver of the platform. It orchestrates group formation, member tracking, discount unlocking, and payment processing.

- User selects a product and clicks **Start a Team** or **Join Existing Team**
- System generates a unique Group ID and shareable invite URL
- A 24-hour countdown timer is started at the moment of group creation
- Each member who joins via the invite link is added to the group roster
- Progress bar and member count update in real time on the product page
- When the target member count is reached, group status changes to **Complete**
- Payment pre-authorization is converted to a charge for all members simultaneously
- If the group does not fill within 24 hours, all pre-authorizations are released
- Group creator may extend the timer once using the **Extend Timer** option
- Group creator may cancel at any time — all pre-authorizations are released

4.3 Timer & Expiry Logic

Each group deal is time-bound to create urgency and drive faster member recruitment.

- Default group timer: 24 hours from group creation
- Countdown displayed live on the group status page (HH:MM:SS format)
- Timer expiry message: *'The price resets to [original price] when timer hits zero'*
- One-time timer extension per group (creator only): resets timer by a configurable duration
- On expiry without group completion: all payment pre-authorizations are auto-voided
- Expired groups are archived in the Admin Panel for manual review
- Admin can manually mark an incomplete group as 'Complete' to override expiry

4.4 WhatsApp Integration

WhatsApp is the primary growth lever for group recruitment. The platform integrates WhatsApp Business API to enable one-tap sharing, automated messages, and group status notifications.

- **Invite Sharing:** Pre-filled WhatsApp message with product name, group discount, and invite link
- **Join Notification:** Automatic message to group creator when a new member joins
- **Group Complete Alert:** Blast to all members confirming group success and payment capture

- **Expiry Warning:** Reminder to all members when only 1 hour remains on the timer
- **Timer Extended:** Notification to all members when the creator extends the countdown
- **Group Cancelled:** Notification with confirmation that payment hold has been released

4.5 Pricing & Discount Tiers

Each product listed on the platform must have a defined pricing grid across group sizes. The seller sets the prices per tier; the platform enforces and displays them.

Group Size	Tier Label	Discount	Behaviour
1 (Solo)	Solo Price	0%	Full price, no group required
2 Members	Duo Deal	~10%	Unlocked when 2nd member joins
3 Members	Trio Deal	~15%	Unlocked when 3rd member joins
5 Members	Team Deal	~25%	Unlocked when 5th member joins
10 Members	Best Deal	~33-40%	Max discount, limited slots, highest saving

4.6 Payment Gateway & Pre-Authorization

- **Pre-Authorization Hold:** Payment is authorized but not captured at checkout; capture occurs only on group completion
- **Auto-Release:** Authorization is voided automatically if the group expires or is cancelled
- **Supported Methods:** UPI (Google Pay, PhonePe, BHIM), Credit/Debit Cards (Visa, Mastercard, Amex, RuPay), Cash on Delivery (for selected pincodes)
- **Gateway:** Razorpay or PayU — to be finalized based on client preference; API costs excluded from project scope
- **Refund Handling:** Automated refund trigger for any group that fails to complete
- **Invoice:** System-generated order invoice with GST breakdown post-purchase

4.7 Seller Panel

- Seller registration and profile management (name, bank details, GSTIN)
- Product listing: name, description, images, category, SKU, and stock
- Group pricing configuration: set price per tier (2, 3, 5, 10 members)
- Order management: view incoming orders, update fulfillment status
- Shipping tracking: add courier name and tracking number per order
- Sales dashboard: revenue, active groups, completed orders
- Seller approval required by Admin before going live

4.8 Admin Panel

- Dashboard: total active groups, completed deals, revenue, new users (today/week/month)
- Seller management: approve, suspend, or remove sellers; view seller performance

- Product management: review and approve new product listings, edit/delete products
- Order management: full order history across all sellers with status filters
- Group management: view all active/expired/completed groups; manually mark incomplete groups as complete
- Category management: add, edit, delete product categories
- Coupon/promo management: create platform-wide discount codes
- Commission configuration: set commission percentage per category or seller
- Notification logs: view WhatsApp and push notification delivery status
- User management: view, ban, or remove registered users

4.9 Notification System

Event	Channel	Recipient
New member joined group	WhatsApp + Push	Group Creator
Group completed (target reached)	WhatsApp + Push	All Group Members
1 hour left on timer	WhatsApp + Push	All Group Members
Timer extended by creator	WhatsApp + Push	All Group Members
Group cancelled by creator	WhatsApp + Push	All Group Members
Group expired (not completed)	Push	All Group Members
Order confirmed	WhatsApp + Email	Buyer
Order shipped	WhatsApp + Email	Buyer
Order delivered	Push + Email	Buyer
New seller registration	Email	Admin
New product pending approval	Email	Admin

5 Non-Functional Requirements

Performance	Page load time under 3 seconds; real-time group status updates without full page reload
Scalability	Architecture supports concurrent active groups with no degradation in response time
Security	SSL/TLS encryption, input sanitization, OWASP Top 10 compliance, secure payment token handling
Availability	99.5% uptime SLA; deployment on cloud infrastructure with auto-recovery
Compatibility	Responsive design supporting Chrome, Safari, Firefox on desktop and mobile browsers

Data Privacy	PII (names, addresses, payment info) stored encrypted; compliant with applicable data protection norms
Audit Trail	All payment events, group state changes, and admin actions logged with timestamps

6 Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	Web UI — responsive, component-driven interface
Mobile App	React Native	Cross-platform mobile apps (iOS & Android)
Backend	Node.js + Express.js	REST APIs, business logic, group engine
Database	PostgreSQL	Relational data: users, products, groups, orders
Cache	Redis	Real-time group state, session management, timers
File Storage	AWS S3 / Cloudinary	Product images, seller assets
Authentication	JWT + OTP (SMS/WhatsApp)	Secure login, session management
Payments	Razorpay / PayU	UPI, card, pre-auth, COD orchestration
Notifications	WhatsApp Business API + FCM	Group alerts and push notifications
Hosting	AWS / DigitalOcean	Scalable cloud deployment, CDN
Version Control	GitHub	Source code management, CI/CD pipelines

7 Third-Party APIs & Integrations

The following third-party services are required for platform operation. Their subscription/usage costs are **not included** in the project price and are to be borne by the client separately.

API / Service	Provider	Usage	Cost Note
Payment Gateway	Razorpay / PayU	UPI, cards, pre-auth, refunds	~2% per transaction
WhatsApp Business API	Meta / Twilio	Invite sharing, group notifications	Per-message pricing
SMS OTP	MSG91 / Twilio	User login OTP verification	Per-SMS pricing
Cloud Storage	AWS S3 / Cloudinary	Product & seller image hosting	Pay-as-you-go
Push Notifications	Firebase (FCM)	Web & mobile push alerts	Free tier available
Maps / Pincode API	Google Maps API	COD pincode eligibility check	Per-request pricing

Email Service	SendGrid / Mailgun	Order confirmations, invoices	Free tier available
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Third-Party Charges — Client Responsibility

All third-party API subscriptions, platform fees, and usage-based charges listed in Section 7 are **strictly excluded** from the ArgosMob project price of **INR 70,000 + 18% GST**. These costs are to be procured, subscribed to, and funded entirely by the client (Sahil Srivastava) before or during the integration phase of development.

Service	Estimated Monthly Cost	Billing Model	Who Pays
Razorpay / PayU (Payment Gateway)	~2% per successful transaction	Transaction fee	Client
WhatsApp Business API (Meta / Twilio)	Rs. 0.40 – Rs. 1.00 per message	Per-message	Client
SMS OTP — MSG91 / Twilio	Rs. 0.15 – Rs. 0.25 per OTP	Per-SMS	Client
AWS S3 / Cloudinary (Storage)	Rs. 500 – Rs. 2,000 / month	Pay-as-you-go	Client
Firebase FCM (Push Notifications)	Free (up to quota)	Free tier	Client
Google Maps / Pincode API	Rs. 0.005 – Rs. 0.02 per request	Per-request	Client
Cloud Hosting (AWS / DigitalOcean)	Rs. 1,500 – Rs. 5,000 / month	Monthly subscription	Client
SendGrid / Mailgun (Email)	Free up to 100 emails/day	Free / Paid tier	Client

IMPORTANT NOTE

ArgosMob Tech & AI Pvt Ltd will provide technical integration of all third-party APIs listed above as part of the project scope. However, the client is solely responsible for:

- 1. Creating and maintaining accounts with each third-party provider.
- 2. Paying all subscription, usage, and transaction fees directly to providers.
- 3. Providing valid API keys and credentials to ArgosMob during the integration phase.
- 4. Ensuring sufficient credits/balance in each service account to avoid integration delays.

Delays caused by unavailability of third-party credentials or insufficient account balance will not impact the project delivery timeline commitment by ArgosMob.

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Project Timeline & Milestones

The project is estimated at **20-25 business days** from the date of project kick-off and advance payment receipt. The schedule below represents a typical sprint breakdown.

Phase	Deliverables	Duration
Phase 1 Discovery & Design	Wireframes, UI/UX designs (all pages), database schema, architecture sign-off	Days 1–5
Phase 2 Core Development	User auth, product listings, seller panel, group buying engine, timer logic	Days 6–12
Phase 3 Integrations	WhatsApp API, payment gateway (pre-auth + capture), push notifications, COD logic	Days 13–17
Phase 4 Admin Panel & QA	Admin dashboard, manual group override, end-to-end testing, bug fixes	Days 18–21
Phase 5 Deployment & Handover	Cloud deployment, SSL configuration, UAT walkthrough, documentation, code delivery	Days 22–25

9 Payment Schedule

Milestone	Trigger / Condition	Base (INR)	GST 18%	Payable (INR)
Milestone 1 Advance — 40%	Project kick-off, prior to design commencement	28,000	5,040	33,040
Milestone 2 Mid-Delivery — 30%	Core development complete; group engine & seller panel delivered	21,000	3,780	24,780
Milestone 3 Final — 30%	Full platform delivered, tested, and deployed to production	21,000	3,780	24,780
TOTAL	—	70,000	12,600	82,600

Payment Notes:

- All payments are to be made via bank transfer or UPI to ArgosMob Tech & AI Pvt Ltd
- GST Invoice will be issued for each milestone payment as per applicable tax norms
- Work on the next phase commences only upon receipt of the corresponding milestone payment
- Third-party API and cloud hosting costs are not included and are payable by the client

10 Terms & Conditions

1	Intellectual Property	Upon receipt of final payment, full intellectual property rights for the custom codebase transfer exclusively to the client. Open-source libraries used remain subject to their respective licences.
2	Warranty & Support	ArgosMob Tech & AI Pvt Ltd provides 90 days of complimentary post-delivery support covering bug fixes directly attributable to original development work. New features or scope changes will be quoted separately.
3	Change Requests	Any change in scope after BRD sign-off will be assessed and quoted as a Change Request (CR). Approved CRs may affect timeline and cost.
4	Confidentiality	Both parties agree to keep all project details, pricing, and business strategies strictly confidential for a period of two (2) years from the project completion date.
5	Limitation of Liability	ArgosMob Tech & AI Pvt Ltd's liability shall not exceed the total project value paid. The company is not responsible for losses arising from third-party service outages, payment gateway failures, or client-side operational decisions.
6	Governing Law	This agreement shall be governed by the laws of India. Any disputes shall be resolved through arbitration in Greater Noida, Uttar Pradesh.
7	Project Commencement	Development commences upon receipt of Milestone 1 payment and written sign-off on this BRD by the client.

Agreement & Sign-Off

By signing below, both parties confirm they have read, understood, and agree to the requirements, scope, timeline, and payment terms as defined in this document.

For ArgosMob Tech & AI Pvt Ltd	For Client — Sahil Srivastava
Authorised Signatory ArgosMob Tech & AI Pvt Ltd	Client Signature Sahil Srivastava
Date: _____	Date: _____